# Rohan Batra

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## SUMMARY

Computer Science undergraduate (UPES, 2027) who builds and documents real systems — from Proxmox and Linux homelab infrastructure to containerized Python services on AWS and agentic AI tooling with evaluation and security guardrails.

## SKILLS

**Languages:** Python, TypeScript, JavaScript, Shell, C, C++, Swift, TeX
**Infrastructure:** Docker, Helm, Nginx, Proxmox, Terraform, ADB, Evolution X, Fastboot, Garuda Linux, Kali NetHunter
**Networking:** DHCP / NAT, DNS, IP Allocation & Subnetting, pfSense, Pi-hole, NetBird
**Backend:** Flask, FastAPI, Pydantic, Uvicorn, Alembic, SQLAlchemy, Android SDK, Hugo, Microservices, REST APIs
**Data:** MongoDB, Redis, PostgreSQL, ChromaDB, MySQL, DuckDB, DuckLake
**AI/LLM:** Claude, Anthropic API, LangChain, Ollama, Agent Evals & Guardrails, Agent Hooks, Agent Skills, Anthropic Messages API
**Security:** Gitleaks, HashiCorp Vault, Semgrep, Trivy, age encryption, AppArmor, Burp Suite, Cerbos
**DevOps:** GitHub Actions, MkDocs, Obsidian, Termux, cargo-deny, cargo-llvm-cov, cargo-mutants
**Frontend:** CSS, Next.js, React, Tailwind CSS, HTML, Dart

## PROJECTS

**Paridyum — Independent Multi-Tenant Platform** *Rust, OpenTofu, Ansible, Talos Linux, Kubernetes, Cilium, Next.js, PostgreSQL*
Self-directed systems project: built solo from an empty directory to a running five-node Kubernetes cluster on self-hosted metal with a public internet edge.

- Built a **specification-as-software governance system**: 251 versioned artifacts (constitution, architecture decision records, standards, RFCs) with machine checks mapped outward to **NIST SSDF, CIS v8.1 and NIST CSF 2.0**, a multi-axis test-type taxonomy, and a precedence order in which existing code is the lowest authority.
- Delivered the full infrastructure path: hardened bare metal (AppArmor, nftables, confined QEMU/libvirt) through **Proxmox and OpenTofu** to **Talos Linux and Kubernetes** with Cilium as CNI and kube-proxy replacement, L2-announced load-balancer VIPs, and etcd snapshots that assert they captured something.
- Wrote **31,000+ lines of Rust** across two production tools — hexagonal architecture, trait-based abstraction, injectable clock adapters, and three independent SHA-256 hash chains for tamper-evident audit records.
- Engineered a **governable AI development harness**: a Claude Code plugin whose PreToolUse hooks make violating a locked architectural decision impossible rather than discouraged, with class-filtered context injection and append-only research provenance.

**AutoTriage — Autonomous Vulnerability Triage & Remediation Agent** *Python, Claude Agent SDK, Semgrep, Trivy, Gitleaks, Pydantic, Terraform*
An agent that triages real scanner findings end-to-end, with evals and human-in-the-loop guardrails.

- Built an autonomous vulnerability-triage agent on the Claude Agent SDK that ingests Semgrep, Trivy, and Gitleaks findings, reasons about severity and business impact, and files tickets, assigns owners, and drafts remediation PRs.
- Enforced human-in-the-loop guardrails via Pydantic validators — confidence-threshold escalation, fail-closed on malformed output, and prompt-injection defenses that treat scanner output as untrusted input.
- Built a fix-validation loop that applies proposed patches to an isolated copy of the code and re-runs the scanner, accepting a fix only when the finding is verifiably resolved with no regression.
- Designed an eval harness scoring triage decisions against a labelled ground-truth set: **100% verdict accuracy** across 44 live findings triaged into 24 tickets and 20 human escalations, backed by 68 tests at 89% coverage.

**UPES-ECS — Campus Emergency Communication System** *Juniper SRX/EX/Mist, Asterisk/PJSIP, FastAPI, QEMU, PowerShell*
LAN-only emergency calling on HPE Juniper networking. Presented to Team HPE at their Bengaluru campus and at UPES, July 2026.

- Built a **LAN-only** campus emergency calling system (one-number 111 hotline) on Asterisk/PJSIP requiring no internet, SIM, or cloud — designed, deployed and operated end-to-end on an **HPE Juniper** networking backbone.
- Configured the enterprise fabric by name: **Juniper SRX300/320** as the guarded edge (firewall, DHCP, NTP, LAN-only boundary), a **Virtual Chassis of two EX2300-C-12P** PoE switches carrying the voice network, and two **Mist AP32** Wi-Fi 6 access points for campus-wide softphone reach.
- Engineered the voice fabric for zero-touch operation and self-healing: **LLDP-MED** auto-provisioning of voice VLAN, PoE class and DSCP; strict-priority **CoS for RTP**; **DHCP-snooping with Dynamic ARP Inspection** to protect the failover path; and Junos **event-options** policies that power-cycle a hung AP without human intervention. Budgeted PoE against the switch's real 124 W ceiling.
- Designed **offline caller location by switch port** (IP to MAC to port to room) so the network identifies where a 111 call originated with no cloud or location service, and fed live switch/AP state to the Operations Console over

**NETCONF**.

- Operated it as a production system where downtime is a safety defect — active/standby high availability, health checks, load testing, and incident root-cause analysis. Three documented RCAs: a serialized single-threaded proxy, a NAT media-address fault dropping every call at 32 seconds, and leaked channels with no teardown path.
- Developed a FastAPI status/control API exposing real-time queue, call, and system state to an operations console and live LED-TV wallboards, provisioned end-to-end via a single PowerShell installer.
- Implemented multilingual IVR across **44 languages** with Piper TTS voice prompts and an offline panic-coach fallback, gated by CI (Ruff, ShellCheck, PSScriptAnalyzer, gitleaks). Presented to **Team HPE** at HPE's Bengaluru campus (15 Jul 2026) and demonstrated live at UPES (20 Jul 2026).

**Gap Hunter — Offline Policy Gap Analyzer** *Python, LangChain, Ollama, Pydantic, Electron, GitHub Actions*
HACKIITK 2026 finalist — top 0.4% of 1,300+ teams. Offline AI policy analysis against NIST CSF 2.0.

- Built an offline AI-powered policy gap analyzer mapping security policies against **NIST CSF 2.0** and the CIS MS-ISAC framework, producing evidence-grounded gap reports, revised policy drafts, and remediation roadmaps — with no external API dependency, so sensitive policy text never leaves the host.
- Designed a multi-stage pipeline (PDF/DOCX ingestion to hybrid retrieval combining vector, sparse, and reranking to gap analysis to policy revision) orchestrated via LangChain with Pydantic-validated structured outputs to ground LLM reasoning and reduce hallucination.
- Shipped a cross-platform Electron desktop app with local Ollama runtime integration and a **702-test suite** at 97.7% pass rate spanning unit, integration, property-based, and adversarial tests.
- Placed **top 0.4% of 1,300+ teams** and 9,000+ participants (C3iHub, IIT Kanpur Cybersecurity); one of 24 finalist teams invited to campus.

## EXPERIENCE

**Hardware & WireGuard Research Intern** *2026-07 – Present*
Exaflair Technologies Pvt. Ltd.

**Platform Engineering Intern** *2025-11 – 2026-07*
Eve

- Founded the **Eve Gateway Stack** on APISIX and etcd — central routing, JWT authentication, rate limiting, CORS, and Prometheus metrics — establishing the platform's central trust and routing layer, with gateway-level tenant isolation, trusted tenant/user header injection, and Cerbos-based authorization rules.
- Founded and led the **Eve Data Management Service**: a FastAPI microservice with source/destination/connection APIs, SQLModel models, Alembic migrations, an Airbyte Config API client, sync orchestration, webhook routes, health checks, and Kubernetes/EKS deployment.
- Implemented **schema-per-tenant multi-tenancy** using PostgreSQL search_path switching, with tenant context propagation and tenant-aware service and database isolation.

**Technical Head** *2025-09 – Present*
Google Developer Groups On Campus (GDGOC), UPES Dehradun

- Led and delivered technical workshops for **100+ participants** across cloud, system design, and full-stack development.
- Managed cross-functional teams executing production-grade initiatives.

**Linux Support & Community Volunteer** *2023-09 – 2025-04*
LinuxWale, Dehradun

- Onboarded **10+ students** to Linux — distribution selection, dual-boot setup, and driver/hardware troubleshooting across varied hardware.
- Wrote shell scripts to automate post-install configuration and routine setup tasks.

## EDUCATION

**UPES Dehradun** — B.Tech (Hons.) Computer Science Engineering, Cyber Security *2023-08 – 2027-08 · CGPA 8.1/10*

## ACHIEVEMENTS

- **Top 15% — TryHackMe cybersecurity learning platform**
- **Certificate of Appreciation — Golden Key Eagles Women Empowerment Center (Sep 2024)**
- **Certificate of Appreciation — HQ Uttarakhand Sub Area (Aug 2024)**
- **Certificate of Appreciation — Golden Key Tuskers, AWWA Week 2024 (Aug 2024)**
- **Semifinalist — Eureka! Junior 2022–23, E-Cell IIT Bombay (Issued Nov 2022; Credential ID: ecell2021)**
- **Semifinalist — Eureka! Junior 2021–22, E-Cell IIT Bombay (Issued Dec 2021; Credential ID: ecell2021)**